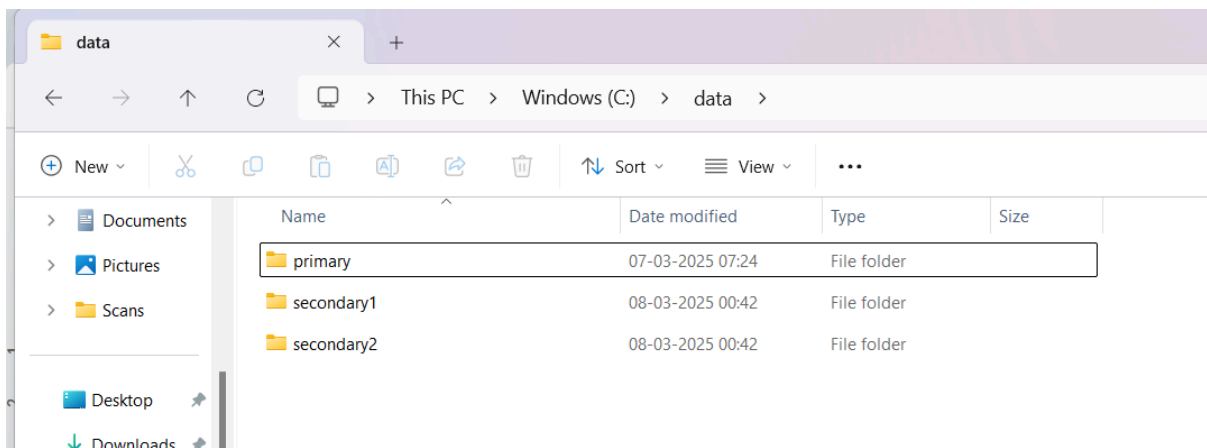
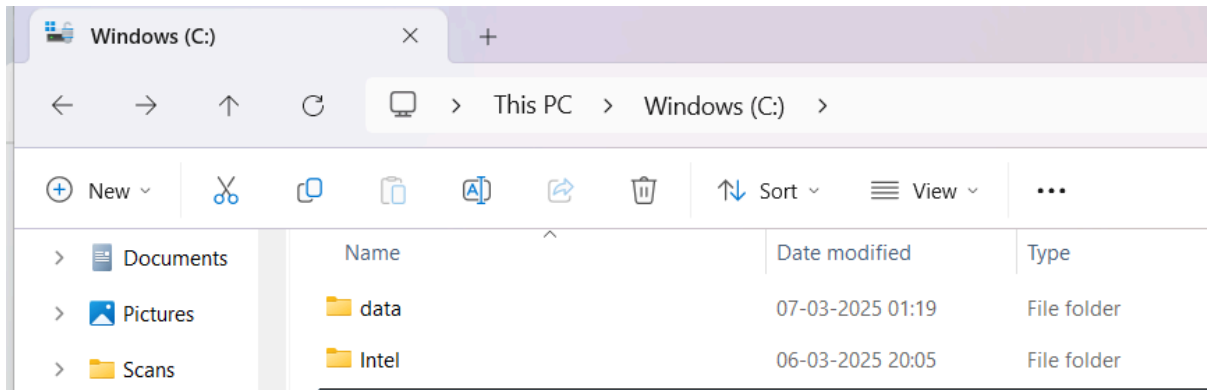


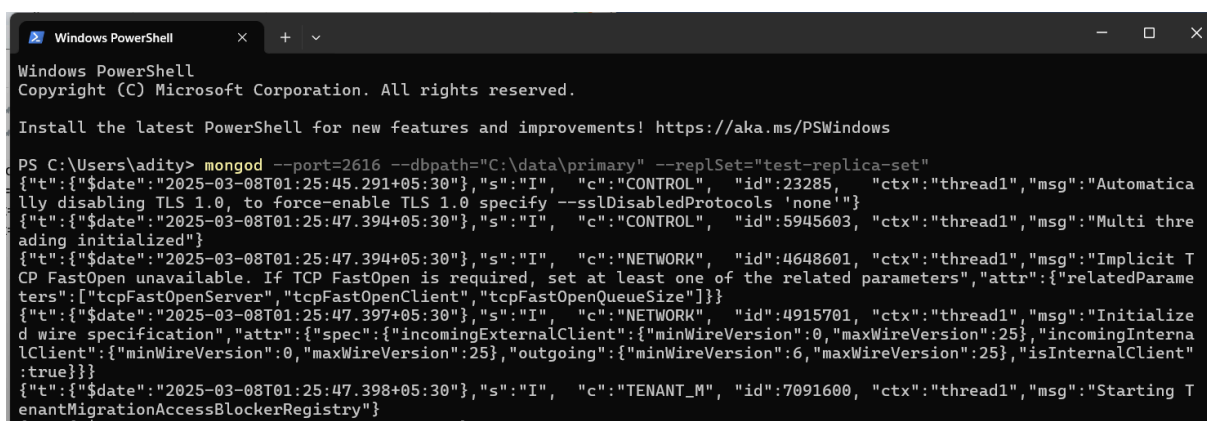
Practical 6 - Replication in MongoDB

- First Create data folder C: location
 - Then create primary , secondary1 , secondary2 folder in it



Now Open windows powershell and start the servers by following commands:

```
mongod --port=2616 --dbpath="C:\data\primary" --replSet="test-replica-set"
mongod --port=2626 --dbpath="C:\data\secondary1" --replSet="test-replica-set"
mongod --port=2636 --dbpath="C:\data\secondary2" --replSet="test-replica-set"
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\adity> mongod --port=2626 --dbpath="C:\data\secondary1" --replSet="test-replica-set"
{"t":{"$date":"2025-03-08T01:26:30.762+05:30"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"thread1","msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2025-03-08T01:26:30.764+05:30"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initialized"}
{"t":{"$date":"2025-03-08T01:26:30.764+05:30"},"s":"I", "c":"NETWORK", "id":4648601, "ctx":"thread1","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"relatedParameters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}
{"t":{"$date":"2025-03-08T01:26:30.764+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":25},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":25},"outgoing":{"minWireVersion":6,"maxWireVersion":25},"isInternalClient":true}}}
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\adity> mongod --port=2636 --dbpath="C:\data\secondary2" --replSet="test-replica-set"
{"t":{"$date":"2025-03-08T01:27:13.124+05:30"},"s":"I", "c":"CONTROL", "id":23285, "ctx":"thread1","msg":"Automatically disabling TLS 1.0, to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2025-03-08T01:27:13.130+05:30"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initialized"}
{"t":{"$date":"2025-03-08T01:27:13.131+05:30"},"s":"I", "c":"NETWORK", "id":4648601, "ctx":"thread1","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set at least one of the related parameters","attr":{"relatedParameters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}
{"t":{"$date":"2025-03-08T01:27:13.133+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":25},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":25},"outgoing":{"minWireVersion":6,"maxWireVersion":25},"isInternalClient":true}}}
```

```
mongosh mongodb://localhost
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\adity> mongosh --host="localhost:2616"
Current Mongosh Log ID: 67cb4fbec7579e98cbfa4213
Connecting to:      mongodb://localhost:2616/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.4.0
Using MongoDB:      8.0.5
Using Mongosh:       2.4.0
mongosh 2.4.2 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2025-03-08T01:25:47.493+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2025-03-08T01:25:47.493+05:30: This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bind_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning
-----
```

```
test> rs.initiate()
{
  info2: 'no configuration specified. Using a default configuration for the set',
  me: 'localhost:2616',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp{ t: 1741377525, i: 1 },
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp{ t: 1741377525, i: 1 }
}
```

```
test-replica-set [direct: primary] test> rs.add({host:"localhost:2626"})
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1741460591, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1741460591, i: 1 })
}
```

```
test-replica-set [direct: primary] test> rs.add({host:"localhost:2636"})
{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1741460623, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1741460623, i: 1 })
}
```

```
test-replica-set [direct: primary] test> rs.status()
{
  set: 'test-replica-set',
  date: ISODate('2025-03-08T19:04:14.697Z'),
  myState: 1,
  term: Long('2'),
  syncSourceHost: '',
  syncSourceId: -1,
  heartbeatIntervalMillis: Long('2000'),
  majorityVoteCount: 2,
  writeMajorityCount: 2,
  votingMembersCount: 3,
  writableVotingMembersCount: 3,
  optimes: {
    lastCommittedOpTime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    lastCommittedWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    readConcernMajorityOpTime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    appliedOpTime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    durableOpTime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    writtenOpTime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    lastAppliedWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastDurableWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastWrittenWallTime: ISODate('2025-03-08T19:04:07.001Z')
  },
  lastStableRecoveryTimestamp: Timestamp({ t: 1741460630, i: 1 }),
  electionCandidateMetrics: {
    lastElectionReason: 'electionTimeout',
    lastElectionDate: ISODate('2025-03-08T19:00:14.384Z'),
    electionTerm: Long('2'),
    lastCommittedOpTimeAtElection: { ts: Timestamp({ t: 0, i: 0 }), t: Long('-1') },
    lastSeenWrittenOpTimeAtElection: { ts: Timestamp({ t: 1741378686, i: 1 }), t: Long('1') },
    lastSeenOpTimeAtElection: { ts: Timestamp({ t: 1741378686, i: 1 }), t: Long('1') },
    numVotesNeeded: 1,
    priorityAtElection: 1,
    electionTimeoutMillis: Long('10000'),
    numCatchUpOps: Long('0'),
    newTermStartDate: ISODate('2025-03-08T19:00:16.807Z'),
    wMajorityWriteAvailabilityDate: ISODate('2025-03-08T19:00:16.910Z')
  },
}
```

```
members: [
  {
    _id: 0,
    name: 'localhost:2616',
    health: 1,
    state: 1,
    stateStr: 'PRIMARY',
    uptime: 253,
    optime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeDate: ISODate('2025-03-08T19:04:07.000Z'),
    optimeWritten: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeWrittenDate: ISODate('2025-03-08T19:04:07.000Z'),
    lastAppliedWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastDurableWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastWrittenWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    syncSourceHost: '',
    syncSourceId: -1,
    infoMessage: '',
    electionTime: Timestamp({ t: 1741460414, i: 1 }),
    electionDate: ISODate('2025-03-08T19:00:14.000Z'),
    configVersion: 7,
    configTerm: 2,
    self: true,
    lastHeartbeatMessage: ''
  },
  {
    _id: 3,
    name: 'localhost:2626',
    health: 1,
    state: 2,
    stateStr: 'SECONDARY',
    uptime: 63,
    optime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeDurable: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeWritten: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeDate: ISODate('2025-03-08T19:04:07.000Z'),
    optimeDurableDate: ISODate('2025-03-08T19:04:07.000Z'),
    optimeWrittenDate: ISODate('2025-03-08T19:04:07.000Z'),
    lastAppliedWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastDurableWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastWrittenWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastHeartbeat: ISODate('2025-03-08T19:04:12.892Z'),
    lastHeartbeatRecv: ISODate('2025-03-08T19:04:12.891Z'),
    pingMs: Long('0'),
    lastHeartbeatMessage: '',
    syncSourceHost: 'localhost:2616',
    syncSourceId: 0,
    infoMessage: '',
    configVersion: 7,
    configTerm: 2
  },
  {
    _id: 4,
    name: 'localhost:2636',
    health: 1,
    state: 2,
    stateStr: 'SECONDARY',
    uptime: 30,
    optime: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeDurable: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeWritten: { ts: Timestamp({ t: 1741460647, i: 1 }), t: Long('2') },
    optimeDate: ISODate('2025-03-08T19:04:07.000Z'),
    optimeDurableDate: ISODate('2025-03-08T19:04:07.000Z'),
    optimeWrittenDate: ISODate('2025-03-08T19:04:07.000Z'),
    lastAppliedWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastDurableWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastWrittenWallTime: ISODate('2025-03-08T19:04:07.001Z'),
    lastHeartbeat: ISODate('2025-03-08T19:04:12.891Z'),
    lastHeartbeatRecv: ISODate('2025-03-08T19:04:13.418Z'),
  },
]
```

Perform CRUD operations in primary node access it using secondary node

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\adity> mongosh --host="localhost:2626"
Current Mongosh Log ID: 67cc954a2b289e430bfa4213
Connecting to:      mongodb://localhost:2626/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.4.0
Using MongoDB:      8.0.5
Using Mongosh:       2.4.0
mongosh 2.4.2 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2025-03-09T00:30:31.829+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2025-03-09T00:30:31.830+05:30: This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bind_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning
-----
```

```
mongosh mongodb://localhost:2626
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\adity> mongosh --host="localhost:2636"
Current Mongosh Log ID: 67cc957b7071715e3afa4213
Connecting to:      mongodb://localhost:2636/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.4.0
Using MongoDB:      8.0.5
Using Mongosh:       2.4.0
mongosh 2.4.2 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2025-03-09T00:30:55.333+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2025-03-09T00:30:55.333+05:30: This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bind_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning
-----
```

```
test-replica-set [direct: primary] test> show dbs
admin      80.00 KiB
config     176.00 KiB
local      460.00 KiB
test-replica-set [direct: primary] test> use student
switched to db student
```

```
test-replica-set [direct: primary] student> db.users.insert({name:"adi",age:12})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('67cc961708f6ac9864fa4214') }
}
test-replica-set [direct: primary] student> db.users.find()
[ { _id: ObjectId('67cc961708f6ac9864fa4214'), name: 'adi', age: 12 } ]
```

```
test-replica-set [direct: secondary] student> db.users.find()
[ { _id: ObjectId('67cc961708f6ac9864fa4214'), name: 'adi', age: 12 } ]
test-replica-set [direct: secondary] student> |
```

```
test-replica-set [direct: secondary] student> db.users.find()
[ { _id: ObjectId('67cc961708f6ac9864fa4214'), name: 'adi', age: 12 } ]
test-replica-set [direct: secondary] student> |
```

```
test-replica-set [direct: primary] student> db.users.insertMany([ { username: "john_doe", age: 25, status: "active" }, { username: "jane_smith", age: 30, status: "inactive" }, { username: "alice_jones", age: 28, status: "active" } ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67cc96f608f6ac9864fa4215'),
    '1': ObjectId('67cc96f608f6ac9864fa4216'),
    '2': ObjectId('67cc96f608f6ac9864fa4217')
  }
}
```

```
test-replica-set [direct: primary] student> db.users.updateOne( { username: "john_doe" }, { $set: { age: 30 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
test-replica-set [direct: primary] student> |
```

```
test-replica-set [direct: secondary] student> db.users.find()
[ { _id: ObjectId('67cc961708f6ac9864fa4214'), name: 'adi', age: 12 } ]
test-replica-set [direct: secondary] student> db.users.find()
[
  {
    _id: ObjectId('67cc961708f6ac9864fa4214'), name: 'adi', age: 12 },
  {
    _id: ObjectId('67cc96f608f6ac9864fa4216'),
    username: 'jane_smith',
    age: 30,
    status: 'inactive'
  },
  {
    _id: ObjectId('67cc96f608f6ac9864fa4217'),
    username: 'alice_jones',
    age: 28,
    status: 'active'
  },
  {
    _id: ObjectId('67cc96f608f6ac9864fa4215'),
    username: 'john_doe',
    age: 30,
    status: 'active'
  }
]
```

```
test-replica-set [direct: primary] student> db.users.deleteMany({})
{ acknowledged: true, deletedCount: 0 }
test-replica-set [direct: primary] student> |
```